A Finite Black-Hole Information Ledger for Observer-Patch Holography

Working draft under issue B19; not a release asset

The OPH collaboration

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Abstract

In the finite observer-patch model, the two commitments whose apparent conflict constitutes the black-hole information problem are separate machine-checked results: within a supplied pointwise-continuous one-parameter star-automorphism class, private dynamics is forced to be unitary conjugation, and the public record channel is an exact uniform mixture of sign unitaries, hence unital and entropy-nondecreasing. This draft assembles those landed results, the exact data-processing and second-law theorems, and the retained conditional rich-fibre packet into one program: the finite information ledger. Its core claim, partly proved and partly a stated target, is that in this model information is never destroyed but exactly relocated, and the public loss is the computable publicization defect $S(\mathcal{E}(\rho)) - S(\rho) = D(\rho \,\|\, \mathcal{E}(\rho))$, nonnegative and vanishing precisely at the public fixed states. A generalized-second-law-shaped monotone, complementarity receipts on the committed net, and a nondegenerate finite KMS receipt are the companion targets. Every statement is finite; no continuum horizon, collapse, Hawking flux, island formula, or evaporating-spacetime Page curve is claimed, and the black-hole reading is a declared finite-model formulation whose physical attachment is owned by separate live issues.

Status and claim boundary

This is a working draft under live issue B19 and is excluded from release assets. Statements labeled *Theorem* are machine-checked in the Lean library at the cited modules. Statements labeled *Target* are the declared obligations of issue B19 and are open statements; nothing in this draft should be read as claiming them. The phrase “black hole” names the shape of the question, not a constructed physical object: the model has no continuum horizon, no gravitational collapse, no Hawking flux, and no evaporation endpoint. The bridge from the finite ledger to physical black-hole statements requires the source-realized clock (E5), the Einstein-branch geometry (F1), and, for any observable, the target-clean preflight (G2) under immutable custody (G1). The existing compact-transient ringdown custody retains its separate frozen payloads; this draft freezes no prediction.

1 The question in finite form

The black-hole information problem sets two commitments against each other: microscopic dynamics that preserves information, and a macroscopic record that looks exactly thermal. In the observer-patch model both commitments are formal objects, and the question becomes finite: given that ambient dynamics on a finite private algebra is unitary, and given that what any observer can hold is a public record produced by a specific channel, account exactly for the information that the record does not carry.

The model supplies sharper versions of both sides than a generic quantum system would.

First, within the supplied pointwise-continuous star-automorphism class, unitarity is not an additional assumption. The Stone-converse packet proves that every pointwise-continuous one-parameter star-automorphism group of a unital finite-dimensional private star subalgebra is blockwise unitary conjugation generated by one self-adjoint Hamiltonian per block. The theorem makes no claim about dynamics outside that class.

Second, the record channel is not a generic decoherence map. For every projective partition, the publicization pinching is exactly the uniform average of the 2^k independently signed block reflections. The channel that produces the public record is therefore an explicit uniform mixture of unitaries: doubly stochastic in the quantum sense, unital, and entropy-nondecreasing, with the mixing unitaries named.

The ledger program turns these two facts, plus the exact thermodynamic theorems of the library, into a single bookkeeping statement: every bit the public record lacks is located, with an exact formula, in the observer-inaccessible coherences of the ambient state.

2 Landed inputs

Each input below is a machine-checked result in the repository, cited by module. None is new to this draft; the draft's contribution is the assembly and the target list of Section 3.

Theorem 2.1 (Forced unitarity; B3 Stone packet)**.** *Every pointwise-continuous one-parameter star-automorphism group of a unital finite-dimensional private star subalgebra containing the ambient identity is blockwise unitary conjugation, generated per Wedderburn block by one time-independent self-adjoint Hamiltonian, unique up to a real scalar.*

Theorem 2.2 (Publicization is a uniform mixture of sign unitaries)**.** *For every supplied k-block projective partition, the pinching map is the uniform average of the 2^k diagonal sign unitaries (`EventAlgebra/RecordMajorization.lean`); averaging only the two global signs does not pinch, so the independence of the signs is load-bearing.*

Theorem 2.3 (Exact data processing and second law)**.** *Every stochastic kernel preserving a faithful stationary reference contracts Kullback–Leibler divergence to that reference (`kl_push_le`), the conditional heat bath obeys the second law (`heatBath_secondLaw`), and the finite fluctuation theorems hold exactly over the rationals (`Thermodynamics/FiniteConditionalRepair.lean`, `Thermodynamics/FluctuationTheorems.lean`).*

Theorem 2.4 (Support boundary for entropy architecture)**.** *The totalized continuous-functional-calculus logarithm sends every projection to zero, so the raw finite trace formula for relative entropy assigns the value zero to an orthogonal pair of density projections whose support inclusion fails (`EventAlgebra/SpectralEntropyBoundary.lean`). A support-aware extended-value layer is therefore mandatory, not stylistic.*

Theorem 2.5 (Conditional rich-fibre regional packet)**.** *The locally pinned fresh-seed source run supplies split-fibre labels at four pairwise-disjoint observer windows. A declared finite adapter constructs a genuinely noncommutative block algebra with a nonunital two-by-two designated-fibre matrix corner at every observer region, elementwise commutation across distinct constructed blocks, computed four-window coverage of the declared top algebra, and unique restriction gluing with its* *reconstruction corollary (`QFT/RichFibreWitness.lean`, `QFT/RichFibreRegionalNet.lean`). The source does not produce the matrix units, state, regional restrictions, or repair, and the corner is neither a tensor factor nor a `TensorSplitReceipt`. E1 closure requires a source-attached operator algebra and a justified factor or local-channel adapter.*

Theorem 2.6 (Finite KMS identity, degenerate inheritance boundary)**.** *The finite algebraic KMS identity holds for Gibbs states of support-graded flows, and the truncation-8 nets inherit KMS structure only degenerately (`QFT/StructuralInheritance.lean`). The nondegenerate continuation on the rich-fibre net is Target 3.6.*

3 The ledger targets

Throughout, ρ is a finite density matrix on the ambient algebra, $\mathcal{E}$ the pinching of a supplied projective partition, S the support-aware von Neumann entropy, and $D(\cdot\,\|\,\cdot)$ the support-aware relative entropy. Targets 3.1–3.6 are the obligations of issue B19; their statements are the intended final forms and may be narrowed by proved counterboundaries, which would themselves discharge the corresponding obligation.

Target 3.1 (Support-aware entropy core)**.** *Exact finite definitions and basic laws for $S(\rho)$ and $D(\rho\,\|\,\sigma)$ with explicit support handling; the pinching identity*

$$S(\mathcal{E}(\rho)) - S(\rho) = D(\rho\,\|\,\mathcal{E}(\rho)) \geq 0,$$

with equality exactly at the public fixed states, where ($\mathcal{E}(\rho) = \rho$); and entropy non-decrease for the sign-unitary mixture specialized from unitality. This core is the subset of the B9 spectral-information chain that the ledger requires; B9 retains the general majorization, maximum-entropy, and H-theorem scope.

Target 3.2 (No-loss theorem)**.** *Under the committed ambient repair dynamics the global state evolves by the forced unitary conjugation of the Stone packet. Ordinary entropy of the state is preserved; when both arguments co-evolve under the same unitary, each explicitly named jointly unitarily invariant quantity is also preserved (including support-aware relative entropy and trace distance when those quantities are defined). The statement quantifies over the committed dynamics class; no toy evolution is substituted and no arbitrary coordinate-dependent divergence is covered.*

Target 3.3 (Ledger identity and relocation)**.** *For the committed record channel, the public entropy gain equals $D(\rho\,\|\,\mathcal{E}(\rho))$; the quantity is nonnegative, vanishes exactly at the public fixed states, and is carried entirely by the off-block coherences that no access cut of the committed net exposes. Information is relocated, never destroyed, and the relocation is exactly quantified.*

Target 3.4 (Generalized-second-law-shaped monotone)**.** *The composition of the cap first law with Kullback–Leibler contraction yields one exact monotone combining record entropy and the capacity ledger under repair, with finite constants and a negative control exhibiting failure of the monotone when unitality of the record channel is dropped.*

Target 3.5 (Complementarity receipts)**.** *Compose the generic E6 access-cut interface with the retained rich-fibre literals and a justified E1 operator/local-channel adapter. Prove a two-observer ledger in which each accessible algebra determines its own record entropy, the declared block-commutation theorem supplies an algebraic locality receipt, and a glued top observable is uniquely reconstructed within the declared restriction system. No physical no-signalling or refinement-natural interpretation follows before that attachment.*

Target 3.6 (Nondegenerate finite KMS receipt)**.** *The modular data of a restricted state on one genuinely noncommutative rich-fibre region satisfies the finite KMS identity at a capacity-linked inverse temperature, nondegenerately. This receipt feeds the E4 thermality row as a landed artifact and does not close E4.*

Target 3.7 (Optional: finite Page-crossover inequality)**.** *Using the exact* 2^k *sign-unitary average in place of Haar integration, an exact finite inequality locating the crossover of accessible-record information for a bipartition of the committed net.*

Remark 3.8 (What a completed ledger would say)*.* With Targets 3.1–3.6 proved, the finite model supports the following exact sentence: ambient unitarity and a thermal public record are simultaneously theorems, and the information the record lacks is neither lost nor paradoxical but equals $D(\rho \,\|\, \mathcal{E}(\rho))$, an explicitly computable functional that the access structure of the net localizes in observer-inaccessible coherences. In this finite model the information question closes without tension. Whether that closure survives the physical attachments is a separate, harder question owned by the live geometry and clock issues.

4 Relation to standard approaches

The Page analysis, the Hayden–Preskill protocol, and the island computations concern continuum or holographic systems with gravitational dynamics; none of that setting is constructed here. The relation is one of shape, stated plainly: the finite ledger realizes, inside a machine-checked model, the position that unitarity and thermality are compatible because thermality is an access statement rather than a dynamical one. The model contributes two elements that are usually assumptions elsewhere: the unitarity of the microdynamics is forced by a converse theorem rather than postulated, and the record channel is an explicitly named mixture of unitaries rather than a generic coarse-graining. What the model does not contribute is any statement about gravitating systems: there is no horizon, no semiclassical limit, and no replica computation, and this draft asserts none.

5 Observables discipline

No observable statement is made or frozen by this draft. Any future black-hole-adjacent observable derived from the ledger, such as saturation-regime relaxation structure or record-capacity signatures in compact-object transients, must pass through the target-clean G2 preflight with declared clock, frame, scale, and readout contracts, and enter custody under G1 before any data comparison. The existing compact-transient ringdown custody, with its separate frozen payloads and exposure history, is the template and remains untouched by this work.

6 Verification surface

The landed theorems cited in Section 2 carry `#print axioms` receipts with only the standard axioms, no admissions, and no native-evaluation shortcut, in the modules named inline. The B19 obligations land under the same discipline, with the support-aware entropy core coordinated with issue B9 so the definitions exist exactly once. The issue tracker entry for B19 is the execution source of truth for the target list; where a target closes as a scoped no-go, this draft will record the counterboundary in place of the claim.

References

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